

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/361088488>

3D Visualization Supporting Situational Awareness of Model-Based System of Systems

Chapter · June 2022

DOI: 10.1007/978-981-19-3610-4_8

CITATIONS

0

READS

131

5 authors, including:



[Xiaochen Zheng](#)

Southern University of Science and Technology

80 PUBLICATIONS 1,689 CITATIONS

[SEE PROFILE](#)



[Lu Jinzhi](#)

Beihang University (BUAA)

122 PUBLICATIONS 1,165 CITATIONS

[SEE PROFILE](#)



3D Visualization Supporting Situational Awareness of Model-Based System of Systems

Yuqian Cao¹, Boyu Xie¹, Jian Wang¹, Xiaochen Zheng², and Jinzhi Lu²(✉)

¹ University of Electronic Science and Technology of China, Chengdu, China

² Swiss Federal Institute of Technology Lausanne, Lausanne, Switzerland

jinzhi.lu@epfl.ch

Abstract. Model-based System of Systems Engineering (MBSOSE) enables to improve the efficiency of the architecture design, and the performance analysis and modeling of highly complex system of systems (SoS). Introducing situational awareness into complex system of systems engineering (SoSE) can facilitate users to understand and analyze the SoS models and make decisions through the situational awareness. However, conventional texts or models cannot be intuitively understood by users through traditional SoSE approach. Therefore, this paper proposes a 3D visualization technology to support situational awareness using MBSOSE. A semantic modeling solution based on KARMA language and Graph-Object-Port-Property-Relationship-Role (GOPRR) ontology is proposed to combine 3D visualization technology with co-simulation to enhance situational awareness through enhanced vision. It can enhance the capabilities of perception, understanding and prediction of decision makers. A case study is conducted to prove the effectiveness of the proposed solution which the braking scene is modeled in an intelligent vehicle SoS using the 3D visualization technology to dynamically co-simulate the braking scene.

Keywords: MBSE · KARMA · 3D visualization · Situational awareness

1 Introduction

System of systems (SoS) is a complex engineering system composed of many engineering systems [1]. With the development of human society, a variety of systems with complex structures, diverse functions and large scales are proposed. Different systems can provide functions that cannot be realized by a single system through cooperation [2].

Traditional System of Systems Engineering (SoSE) is facing several challenges. First of all, a SoS often involves multiple engineering disciplines, several development teams, and different stakeholders. Various domain specific architecture description languages are used to describe the SoS. It may lead to misunderstandings among developers which can further result in unified SoS models. Secondly, the traditional MBSOSE is usually implemented based on the life cycle model [3], according to which the SoS model design is carried out by models first and then the production of the physical prototype is

carried out. This approach may result in the gap between the real products and SoS design because less verification is implemented before prototyping. The iterative development process led by inefficient verification may greatly increase the production cost and time cycle. Finally, decision-support through modeling and simulation is one of the critical tasks for SoS design. Traditional SoSE requires decision makings for different system capability analysis. Technologies, such as situational awareness has been widely used in order to minimize uncertainty, production costs, and eliminate barriers for verifying the SoS models [4].

In the field of MBSoS, situational awareness is a key factor that determines the effectiveness and timeliness of the system which consists of three independent layers: perception layer, understanding layer and prediction layer. The perception layer mainly senses the state, attributes and dynamics of related elements in the environment. The understanding layer is based on the perception layer to understand the meaning of data and clues to the target. The prediction layer predicts the ability of elements in the short term based on the perception layer and understanding layer.

3D visualization is considered as one important technique to present complex and abstract data information with appropriate visual elements and perspectives, which is easy for users to understand, remember and transfer the SoS development information for supporting situational awareness [7]. Combining visualization with situational awareness can solve the problems mentioned above in MBSOSE [5].

- First of all, compared with other formats of situational awareness, visual vision contains the largest amount of dynamic information during SoS operations. 3D Visualization is used to enhance situational awareness, by collecting more information at the perception layer in order to provide support for the understanding layer and prediction layer.
- Second, the introduction of the 3D visualization in the understanding layer enables the system developers to identify and to understand the rational between the environment factors, and the development trend of the elements. It minimizes the understanding deviations of a model caused by inconsistent knowledge of cross-domain developers.
- Finally, the combination of situational awareness and 3D visualization can enhance the functional capabilities of the situational awareness prediction layer [6].
 - On one hand, it can visualize the system design process through the knowledge required for verification. Therefore, the development and design can be constantly adjusted through a more understandable visualization during the verification process. Correspondingly, the production cost and cycle time can be properly controlled with less risks lead by the iterative design.
 - On the other hand, after the integrating MBSOSE, situational awareness can be used in a specific scene to understand the dynamics of each system in a certain time and space.

In this paper, 3D visualization is applied to complex SoSE situation awareness. The data of visualization file is converted into CZML format for static display with 3D models in a developed plugin. Meanwhile, With co-simulation, the situation awareness is implemented with simulations implemented within the SoS model development. The

scene is created based on a 3D visualization model combining the hybrid automata simulation execution. It can help SoS designers realize the integration of the dynamic performances of SoS operations, thus to reduce the complicated SoS development cost and time. At the same time, it is beneficial for the users to make quick decisions in specific scenarios.

The rest of the paper is organized as follows. In the second chapter, related work is discussed, including the existing problems in SoSE, the proposal and development of situational awareness in the sSoSE, and how the current technology realizes SoS modeling and situational awareness. Then, the method of this paper is introduced in Sect. 3. SoS modeling is carried out on MetaGraph 2.0¹ using KARMA language based meta-model library and model for SoS. 3D visualization is integrated into MetaGraph 2.0 for the situational awareness, based on the 3D visualization models which are configured to implement co-simulation for 3D visualization dynamic simulation for specific scenes. In Sect. 4, a case study of auto-braking scene in intelligent vehicle SoS is presented, and the role of the proposed method during automatic driving system process is discussed and analyzed. Finally, the conclusion is given in Sect. 5.

2 Related Work

SoSE mainly experienced three stages: text-based SOSE, MBSOSE and simulation modeling SOSE. There are many problems in text-based SOSE, such as too much text and not easy to understand. Daniele Gianni et al. proposed an ESA Architecture Framework (ESA-AF) in 2012 to support SoSE activities such as Galileo Navigation, Global Environment and Safety Monitoring (GMES) and Space Situational Awareness (SSA) projects [8]. However, MBSOSE using the architecture approach still has problems, such as the inconsistency between the product and the design in the physical integration stage, which increases the cost and production cycle [9]. Thus, Zhang Lin et al. proposed Modeling and Simulation Based System of Systems Engineering (MSBS2E) [1], aiming at modeling the full life cycle of products, digital integration of products through simulation technology for verification, testing and situational awareness of specific scenarios in complex SoS [10].

Situational awareness was first proposed by the US Air Force in the 1980s and used in aerospace research. It mainly consists of three levels: perception, understanding and prediction [11]. In Wikipedia says “*Situational awareness (SA) is the perception of environmental factors and events related to time or space, the understanding of their meaning, and the prediction of their future state*” [12]. In 1988, Endsley proposed that situational awareness is a process of acquiring various elements in the environment from the perspective of time and space, understanding these elements and predicting their future state. Using situational awareness was conducive to eliminating the barriers between different professionals in complex systems [13]. In 1995, Endsley proposed that situational awareness was important for system operation. He proposed a situational awareness model based on human dynamic decision making, which can be used in specific scenarios of complex systems, using the predictive function of situational awareness to support system decision making [14].

¹ <http://www.zkhoneycomb.com/>

In complex SoSE, situational awareness undertakes the functions of removing domain barriers in the design and development stage, supporting simulation verification, and supporting user decision-making through situational awareness in the application stage. The above functions are mainly realized by combining the visualization process of complex system model with situational awareness.

In 1999, Card put forward the information visualization model, which mainly includes original data, data table, visual structure and view [15]. In addition, raw data, data tables, visual structures and views could be divided into data layer, visual view and view interaction to support the visualization process in situational awareness. The three layers of situational awareness are respectively supported by different data in the data layer. The overview view, the detail view, and the coexistence of multiple views in the visual view can be used in the decision stage [16].

Based on the above research, it can be found that situational awareness can help eliminate domain barriers, support simulation verification, and support user decision-making in complex systems. 3D visualization technology can enhance situational awareness by enhancing vision. Therefore, this paper proposes a method that 3D visualization supports situational awareness of MBSoS. This paper mainly adopts a semantic modeling approach based on KARMA language and GOPPRR ontology [17] in Metagraph 2.0 platform to realize system meta-model development and system modeling and a 3D visualization approach for situation awareness. First, KARMA language is used to develop SoS models based on UPDM meta-models [18]. Then system behaviors are defined within the KARMA models in order to support dynamic simulation of the SoS operation scene. During the system simulations, the data of visualization file is first developed and converted into CZML format data for the static display of 3D model in the 3D visualization plugin. At the same time, the variables in the KARMA model are defined to map to the corresponding variables in CZML data used for 3D visualization models, so that the system design and situational awareness based on 3D visual model are integrated to implement co-simulation.

The advantages of the proposed method in this paper are twofold:

1. This paper adopts semantic a modeling approach based on KARMA language and GOPPRR ontology to realize the development of SoS meta-model, SoS modeling, SoS simulation and visualization.
2. 3D visualization is used to be integrated into Metagraph platform and realize situation awareness of complex SoSE by configuring 3D visualization model for simulations.

3 3D Visualization Model Construction for MBSOSE Situational Awareness

3.1 The Overall Framework

As shown in Fig. 1, an integrated framework based on KARMA language and 3D visual simulation for MBSoS situation awareness is proposed. There are three main steps to implement the entire 3D simulation and visualization process:

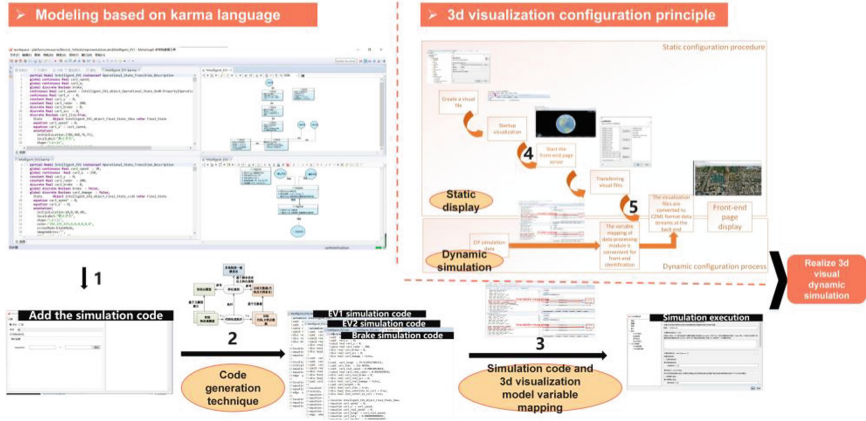


Fig. 1. An integrated framework based on KARMA language and 3D visual simulation for MBSoS situation awareness.

1. Develop SoS models based on KARMA UPDM meta-models. In order to define the SoS formalism, meta-models including 56 graphs are used to define different architectural views of SoS.
2. When developing the brake models for system status definition, variables and equations based on hybrid automata are defined based on KARMA language.
3. After developing the KARMA models, compiling for co-simulation is executed to transform the KARMA models representing system status into execution codes.

After the preparing the co-simulation execution codes, several steps are implemented to develop 3D visualization models and execute the simulation.

4. 3D visual file is developed with 3D models in order to construct the situation awareness scene as shown in Sect. 3.3.
5. The 3D visual models are transformed into CZML codes in order to demonstrate the 3D earth scene.

In order to support the co-simulation between KARMA models and 3D visualization models, a co-simulation configuration is implemented to map the variables generated from hybrid automata simulation to the variables in the 3D visualization scene. Finally, a co-simulation is implemented.

3.2 Co-simulation for 3D Visualization and Hybrid Automata Simulation

We use hybrid automata to describe the behaviors in the system, and the hybrid automata model can be represented by a mathematical model, as shown in Eq. 1:

$$HA = (DSS, TF, FS, VIS, INV, FL, JU, DE) \quad (1)$$

The mathematical model is an 8-tuple. *DSS* presents a limited set of discrete states. *TF* presents transitions between discrete states. *FS* presents a limited set of variables. *VIS* presents a set of initial states. *INV* presents a set of invariants. *FL* presents the continuous

evolution in one state. *JU* presents a set of functions which allocated to each of the *TF*. *DE* presents a set of discrete events.

We set up specific mapping between elements in hybrid automata and elements in KARMA model, define the grammar of hybrid automata through the above mapping rules, synthesize the grammar of hybrid automata and KARMA language, and conduct system modeling based on the synthesized language. The simulation compiler is used to compile the system model describing the system behavior into a.sim file, which is solved by the hybrid automata solution model to generate simulation results, thus supporting dynamic verification.

The data interaction between the hybrid automata simulation process and dynamic 3D visualization process in dynamic simulation is shown in Fig. 2:

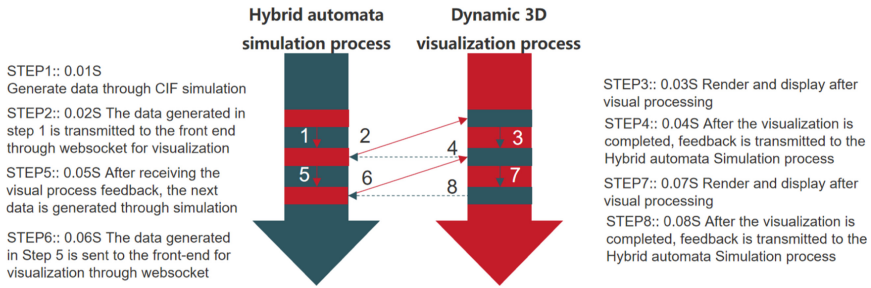


Fig. 2. Hybrid automata simulation process and dynamic 3D visualization process.

The steps are as follows:

1. Generate data through CIF simulation in the hybrid automata simulation process.
2. Transmit the data generated in step 1 to the front end through websocket for visualization.
3. Render and display after visual processing in the hybrid automata simulation process.
4. Transmit feedback to the hybrid automata simulation process after the visualization is completed.
5. Generate the next data through simulation in the hybrid automata simulation process after receiving the visual process feedback,
6. Repeat the process until the end of the simulation.

The co-simulation solution mechanism in the hybrid automata simulation process is described below. The co-simulation system can be regarded as a discrete system composed of several subsystems. In each subsystem, the state variables are shown in Eq. 2, the output equation is shown in Eq. 3. The input and output of each subsystem exchange data at a specified point in time. We calculate the input vector of each subsystem and the output of other subsystems by a coupling formula, as shown in Eq. 4. This coupling formula connects the different subsystems in the co-simulation.

$$x^i = k^i(x^i, u^i, t), x^i(t_0) = x_o^i \quad (2)$$

$$y^i = h^i(x^i, u^i, t), i = I, II, III, N \quad (3)$$

$$u^i = M^i y = [M^{i,I}, \dots, M^{i,i-1}, 0, M^{i,i+1} \dots M^{i,N}] \begin{bmatrix} y^I \\ \vdots \\ y^{i-1} \\ y^i \\ y^{i+1} \\ \vdots \\ y^N \end{bmatrix} \quad (4)$$

i stands for different subsystems, $x^i \in R^{n_x^i}$ is the state variable, $x^i \in R^{n_u^i}$ is the input vector, $y^i \in R^{n_y^i}$ is the output vector, $t \in R$ stands for time, i stands for subsystem, $M^{i,j} \in R^{n_x^i \times n_y^j}$ stands for the matrix element, and can be 0 or 1.

3.3 3D Visualization Model Configuration

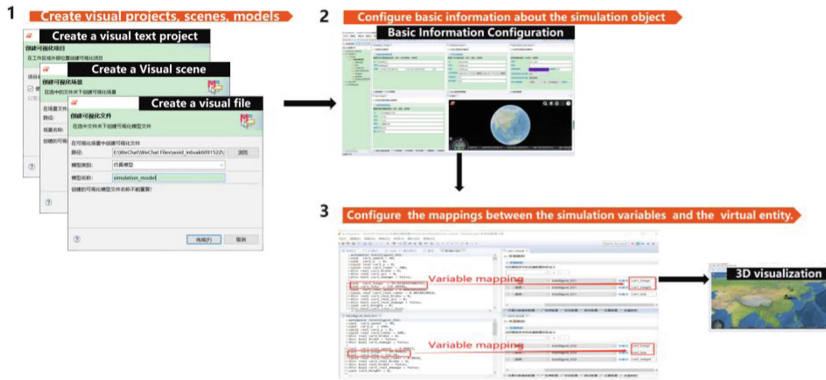


Fig. 3. 3D visualization model configuration process.

This section introduces building a 3D dynamic visualization for a co-simulation code file based on Sect. 3.2. In order to construct 3D dynamic visualization, 3D visual models are developed in a 3D visualization plugin in MetaGraph 2.0 as shown in Fig. 3. The specific steps are as follows:

1. Create a visual scene to represent the visual earth scene.
2. Add static entities as shown in Table 1 in order to construct all the 3D entities in the visualization
3. Configure the mappings between the simulation variables which are generated in each step by hybrid automata simulation solver and the status of each virtual entity.

Table 1. Visual entity for developing visualization scene.

Visual model	Meaning	Basic information
The simulation model	3D visualization models such as car, radar, passenger plane, fighter plane, control center, satellite and warship can be configured for the ontology. The 3D visualization model is associated with the simulation through variable mapping	Emulation object basic configuration
		Billboard configuration
		The tag
		The path configuration
		Location configuration
		Variables mapping
CZML header file	A visual simulation model for connecting transferable information as a visualization of information interaction	CZML Basic information
Radar wave	A visual simulation model for simulating radar scanning waves that can detect objects in the scanning range	Radar scan effect configuration
Information line	A visual simulation model for connecting transferable information as a visualization of information interaction	Information line configuration
Explosion effect	Used to simulate 3D visualizations that explode when entities collide	Explosive effect configuration
Cube	Used for 3D visualization of cubic building models such as buildings and warehouses	Basic configuration
		Cube configuration
Cylinder	Used for 3D visualization of cylindrical buildings or models	Basic configuration
		Cylinder configuration
Billboard	Used for billboard 3D visualization	Basic configuration
		Billboard configuration
Cone	Used for 3D visualization of cone models	Basic configuration

(continued)

Table 1. (continued)

Visual model	Meaning	Basic information
		Cone configuration
Polyhedron	Used for 3D visualization of polyhedral objects	Basic configuration
		Polyhedron configuration
Sphere	Used for 3D visualization of ball-shaped objects	Basic configuration
		Sphere configuration
Line	Used for 3D visualization of linear objects	Basic configuration
		Line configuration
Dot	Used for 3D visualization of point objects	Basic configuration
		Dot configuration

The 3D visualization model in Table 1 includes dynamic model and static model. The dynamic model means that the model state will change over time, and the model state variables need to be bound with the variables in the simulation code to carry out dynamic simulation, including simulation model, radar wave, information line and explosion effect. The model state of a static model remains constant during the simulation process and is usually used to simulate the environment of a simulation scenario. The static model includes cube, cylinder, billboard, cone, polyhedron, sphere, line and dot.

- After the basic information of the visualization model is configured, we can transfer the CZML files to the visualization process or enter the 3D display page by dragging and dropping the simulation file.

Therefore, the 3D visualization configuration has the following advantages. On the one hand, by configuring the 13 visualization models shown in Table 1, entities can be simulated realistically. On the other hand, the configuration of visual model and the writing of hybrid automata simulation code can be carried out simultaneously, and finally associated the two processes by variable mapping, so that we can improve the efficiency of 3D visualization model configuration.

4 Case Study

In this case, KARMA meta-model library and models are used for SoS modeling in MetaGraph 2.0. 3D visual simulation is used to visualize the auto-braking scene which is executed through co-simulation from KARMA models. In this chapter, the problem statement is first introduced, then KARMA models to represent SoS formalism. Finally, configuration of 3D visualization and co-simulation are introduced to support the dynamic situation awareness of the auto-braking.

4.1 Background

In the case study, two vehicles are moving to the same direction on the road as shown in Fig. 4A. When the *car_1* radar detects ahead road barriers, *car_1* will brake for deceleration; at the same time, when the radar detector of *car_2* detects that the distance from *car_1* is less than a certain value, *car_2* will also brake according to the moving state of *car_1*. At the same time, information is transmitted between vehicles, satellites, signal towers and radars.

According to the auto-braking scene, the entire workflow includes, a total of nine operational tasks as shown in Fig. 4B. Firstly, *car_1* radar scanning is open; if there are obstacles ahead, *car_1* will receive the reflected radar signal of the obstacles, and *car_1* request GPS signal; after *car_1* received GPS signals, it sends radar signal and GPS information to *car_2*; and *car_1* confirms that *car_2* has received the information by confirming the radar signal sent by *car_2*. On the other hand, *car_1* sends a signal to the control center, which then sends a message to *car_1* after the control center processed the messages. After receiving the information from *car_2* and the control center, *car_1* will start braking to complete the transformation of the entire combat state.

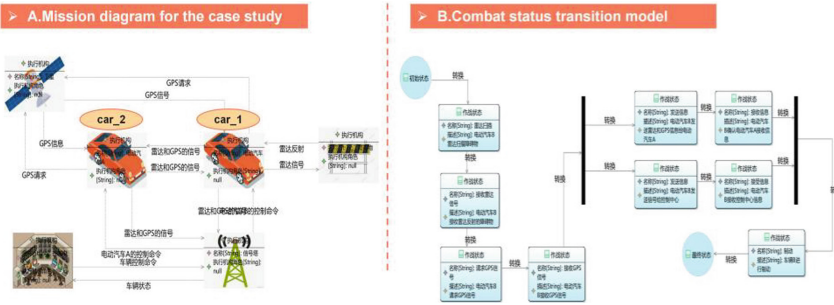


Fig. 4. Auto-braking scene model description

4.2 The SoS Models for System Behaviors Based on Hybrid Automata Simulation

In order to define the SoS operations and system capabilities, SoS models are defined as shown in the Fig. 5: the design operational analysis model, state transition model, and etc. Figure 5C is KAMAR language corresponding to the state transition model, Fig. 5D is a .sim file compiled from the state transition model.

- The design operation analysis model is mainly used to define the use case related to the SoS scenario. It includes three key stakeholders and their system concerns in Fig. 5A.
- The state transition model describes the vehicle's motion state in order to support the auto-braking operation in Fig. 5B. System behaviors are defined based on KARMA language, as shown in Fig. 5C.

- Figure 5C shows the KARMA language corresponding to the state transition model. The KARMA language is based on GOPRR ontology, and the KARMA model is an instance of the graph meta-model, which includes object meta-model, point meta-model, property meta-model, relationship meta-model and role meta-model.
- Figure 5D shows the .sim file of the state transition model compiled by the simulation compiler. In the figure, *automaton* represents the state machine of the state transition model. Each state is marked with *location* in the .sim file, *initial* represents the initial state, *edge* represents the state jump, *when* represents the condition to be met for the state jump, and *goto* indicates the next state of the jump.

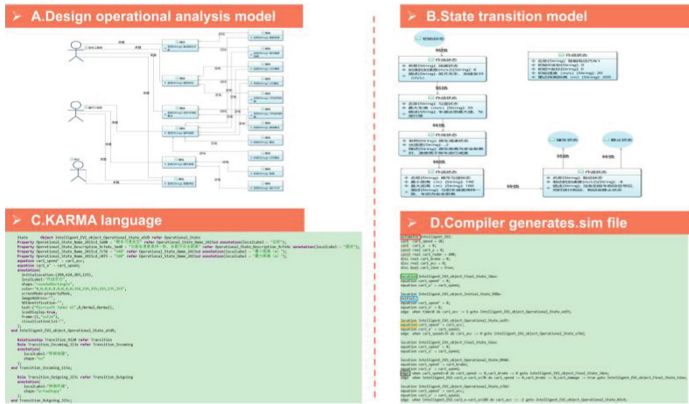


Fig. 5. The SoS models, KARMA language and .sim file

4.3 Configuration for 3D Model Development and Co-simulation

In order to develop 3D models and to configure co-simulation between KARMA models and 3D visualization models, the configuration process is shown in Fig. 6.

When developing the 3D models, visual entities as shown in Table 1 are developed in order to construct the entire situation awareness scenario. 3D visualization model can be divided into dynamic model and static model. Dynamic model includes satellite, control center, vehicle, radar scanning wave and information line. In addition to the configuration of basic information such as size, color, name and initial position, variables of the dynamic model should be bound with corresponding variables in the co-simulation code to support 3D visual dynamic simulation. Information lines should be configured between vehicles and satellites and between vehicles and control center for information interaction. The detailed information interaction process is described in Sect. 4.1. The static model includes cubes and other geometry used to simulate buildings in the braking scene, and its 3D visualization configuration mainly includes size, initial position, color and so on.

Compared with traditional tables or 2D images, the advantages of 3D visualization model in this case are highlighted in the following aspects:

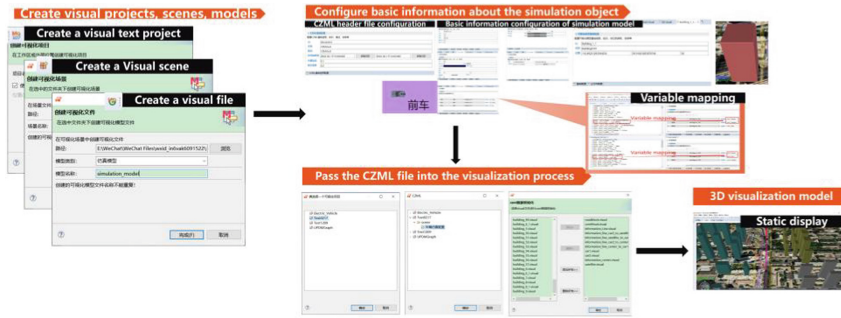


Fig. 6. 3D visualization simulation of configuration process

1. Compared with 2D simulation, only the radius of radar scanning wave can be set. In 3D simulation, the direction, radius, left and right Angle and upper and lower Angle of radar scanning wave can be set, so as to determine a three-dimensional scannable range. This makes the simulation more realistic.
2. During simulation, all models can be viewed from different angles, rather than being limited in a 2D plane, which improves situational awareness.

4.4 3D Visualization Simulation

In this case, the 3D visual simulation site of electric vehicle braking scene is located on the East Fourth Ring Road in Beijing. Through the configuration of the visual model in Sect. 4.3, variable mapping is used to associate the variables of the visual simulation model with the corresponding variables in the simulation code, so as to achieve the purpose of simulation and visual linkage. The 3D visual dynamic simulation of the car braking scene is shown in the video (https://www.bilibili.com/video/BV19L411A7HL?spm_id_from=333.999.0.0). The visual configuration model of this case is shown in Table 2:

Table 2. Auto-braking scene visualization model statistics.

Visual model	Type	Number
The simulation model	The car	2
	Satellite	1
	The control center	1
Information line	—	4
Radar wave	—	1
The cube	—	90

After the above model configuration is completed, the model is transferred to the visualization program for simulation execution. The simulation results and auto-braking scene 3D visualization simulation is shown in the Fig. 7 below:

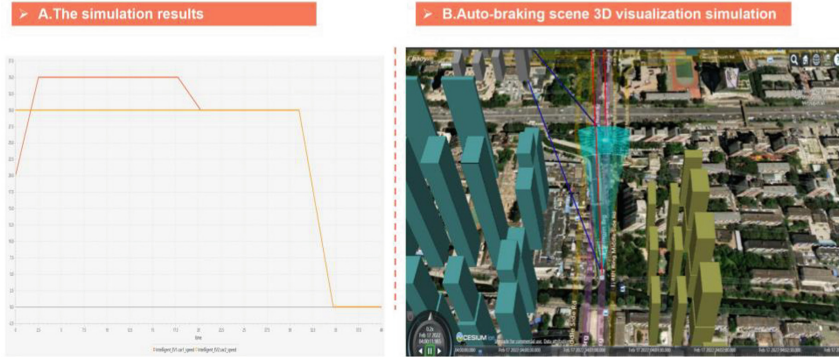


Fig. 7. The simulation results and auto-braking scene 3D visualization simulation

Figure 7A is the speed-time image of two cars, As can be seen from the image, when the speed of *car_2* increases to a certain value, it begins to move at a uniform speed. When the distance between *car_1* and *car_2* is less than a certain value, it slows down to the same speed with car 1. When the distance between car 1 and obstacle is less than a certain value, the two cars slow down at the same time. And Fig. 7B is the auto-Braking scene 3D visualization. Compared to speed-time images, 3D models can quickly enable interdisciplinary experts to understand auto-braking scene; Compared with 2D simulation, 3D dynamic simulation can more restore the real space, enhance the authenticity and detail of the visual level of situation awareness, and help decision makers to make decisions.

4.5 Discussion

From the case study, we find six visualization models including car, satellite, control center, information line, radar wave, cube. They are developed to realize the co-simulation of the auto-braking scene. Through a 40-s simulation, technical developers can have the most intuitive understanding of the auto-braking process without deep professional knowledge. During the visualization, visual efforts, such as information lines and radar waves are integrated with transparent data transmission with physical entities, which are generated from co-simulation. The visualization enhances the visibility of data and knowledge extraction, data interaction and data processing in in order to support MBSOSE situational awareness, and enables users to have a deeper understanding of system dynamics during SoS operations.

3D visualization is used to support MBSOSE situational awareness in the auto-braking scene. It can display the relative position and shape proportion between different vehicle entities which received data from simulations based on KARMA models, which is convenient for users to receive information more intuitively, to understand the system

dynamics, and verify the system capability based on the simulation results. Moreover, 3D visualization can enhance situational awareness from the perspective of enhanced vision, thus improving the accuracy of decision making by SoS developers.

5 Conclusions and Future Work

In this paper, a 3D visualization approach is proposed to support situation awareness for SoS development. KARMA models and 3D visualization models are used to define SoS formalism and simulation awareness scene construction. In order to evaluate the availability of the proposed approach, a case study of auto-braking system is provided. In order to construct the 3D models, six visualization models, including car, satellite, control center, information line, radar wave, cube, which are developed to realize the visual scene definition of the auto-braking scene. Then through co-simulation, 3D visualization models are executed synchronized with the KARMA models. In the future, more engineering cases will be used to verify the method proposed in this paper. Finally, through the visualization, situational awareness is implemented which enables users to have a deeper understanding of system dynamic during the SoS operations.

Acknowledgement. This work was supported by Embedded processor security scanning technology for physically isolated network (2021YFG0338) and Research on security chip for complex electromagnetic environment simulation system (2021YFG0008).

References

1. Lin, Z., Kunyu, W.: System engineering based on modeling simulation. *J. Syst. Simul.* **34**(2), 179 (2022)
2. Topper, J.S., Horner, N.C.: Model-based systems engineering in support of complex systems development. *Johns Hopkins APL Tech. Digest* **32**(1) (2013)
3. Zhang, L.: Applying system of systems engineering approach to build complex cyber physical systems. In: Selvaraj, H., Zydek, D., Chmaj, G. (eds.) *Progress in Systems Engineering*, pp. 621–628. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-08422-0_88
4. Durso, F.T., Hackworth, C.A., Truitt, T.R., et al.: Situation awareness as a predictor of performance for en route air traffic controllers. *Air Traffic Control Q.* **6**(1), 1–20 (1998)
5. Austin, M., Mayank, V., Shmunis, N.: PaladinRM: graph-based visualization of requirements organized for team-based design. *Syst. Eng.* **9**(2), 129–145 (2006)
6. Herter, J., Ovtcharova, J.: A model based visualization framework for cross discipline collaboration in industry 4.0 scenarios. *Procedia CIRP* **57**, 398–403 (2016)
7. Ahram, T., Karwowski, W., Amaba, B., Fechtelkottter, P.: Power and energy management: a user-centered system-of-systems engineering approach. In: Yamamoto, S. (ed.) *HIMI 2013*. LNCS, vol. 8017, pp. 3–12. Springer, Heidelberg (2013). https://doi.org/10.1007/978-3-642-39215-3_1
8. Piaszczyk, C.: Model based systems engineering with department of defense architectural framework. *Syst. Eng.* **14**(3), 305–326 (2011)
9. Mažeika, D., Butleris, R.: MBSEsec: Model-based systems engineering method for creating secure systems. *Appl. Sci.* **10**(7), 2574 (2020)

10. Gianni, D., Ambrogio, A.D., Tolk, A.: Modeling and Simulation-Based Systems Engineering Handbook. Taylor & Francis Group, Boca Raton (2015)
11. Salmon, P., Stanton, N., Walker, G., et al.: Situation awareness measurement: a review of applicability for C4i environments. *Appl. Ergon.* **37**(2), 225–238 (2006)
12. 2019. https://en.wikipedia.org/wiki/Situation_awareness
13. Endsley, M.R.: Design and evaluation for situation awareness enhancement. In: Proceedings of the Human Factors Society Annual Meeting, vol. 32, no. 2, pp. 97–101. Sage Publications, Los Angeles (1988)
14. Endsley, M.R.: Toward a theory of situation awareness in dynamic systems. In: *Situational Awareness*, pp. 9–42. Routledge (2017)
15. Card, M.: *Readings in Information Visualization: Using Vision to Think*. Morgan Kaufmann, San Francisco (1999)
16. Wu, J., Wang, J.: Visual perception Model based on situation Awareness Theory. *Modern Libr. Inf. Technol.* **7**, 9–14 (2010)
17. Ding, J., Reniers, M., Lu, J., et al.: Integration of modeling and verification for system model based on KARMA language. In: Proceedings of the 18th ACM SIGPLAN International Workshop on Domain-Specific Modeling, pp. 41–50 (2021)
18. Lu, J., Wang, G., Ma, J., et al.: General modeling language to support model-based systems engineering formalisms (part 1). In: *INCOSE International Symposium*, vol. 30, no. 1, pp. 323–338 (2020)